

- Controller (WiFi + logic)

Wemos D1 Mini ESP8266

\$1.34

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AliExpress + others

ESP8266-based, Arduino compatible

Built-in WiFi (perfect for your automation setup)

Small footprint for enclosure builds

3.3V logic (important for HX711 + OLED)

Load Measurement System

Load Cells (x4)

4x 50kg Half-Bridge Load Cells

\$14.70

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eBay - williaalfar7

Each: 50kg → combined scale $\approx$ 200kg capacity

Half-bridge → MUST wire all 4 together

Standard bathroom-scale style configuration

Amplifier (Required)

HX711 Load Cell Amplifier

\$7.80

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Display

1.3" SH1106 I2C OLED Display

\$9.55

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Addicore

128×64 resolution

I2C (only 2 wires: SDA + SCL)

Works great with ESP8266 libraries

User Input (Tare / Start Button)

7mm Momentary Push Button (SPST)

\$6.73

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Harfington

Normally open (perfect for GPIO input)

Use internal pull-up on ESP8266